

Tarun Eswar

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EDUCATION

Washington University St. Louis (B.S. Computer Science + Applied Mathematics) Expected YOG: 2027
GPA: 4.0/4.0 (Dean's List)
Relevant Coursework: Multivariable Calculus, Matrix Algebra, Discrete Math, Computer Engineering, Data Structures & Algorithms, C++ Object-Oriented Software Development Laboratory
Activities: WU Google Student Developer Club, Phi Gamma Nu Professional Fraternity, Rock Climbing Team

EXPERIENCE

Fresenius Medical Care, Inc **Lexington, MA**
Software Engineering Trainee June 2025 | present

- Leveraged the CrowdStrike API (python) to automate monthly KPI metrics by working with the Global Head of Cybersecurity which led to an improvement in operational efficiency by 70% for data leaders / owners
- Contributing to spring boot data server by developing unit tests and developing endpoints for Kafka / MongoDB
- Spearheaded ServiceNow MCP server integration with internal tools to optimize time spent on tickets by 43%.

Rakuten Group, Inc **Remote**
Software Engineering Intern/Trainee Jul 2023 | Aug 2023

- Co-developed a QA chatbot (LLMs) to answer company legal questions (95% coverage) with Tiger Team and used quantization methods to optimize Llama2 and Vicuna and store embeddings in a ChromaDB vector.
- Achieved accurate results with response of ~2.5 seconds (12GB GPU tunneling) and created Django QA model API (Express) with dashboard (React) and JWT / Rakuten legal recommendations to protect documents.
- Contributed KPI UI feature components for a production B2B product (Cognitive AIOps) on a three-stage testing workflow with proprietary component libraries. Worked with Redux, Typescript, and Vue framework.

Tufts Financial Group **Medford, MA**
Contract Developer Sep 2022 | Jan 2023

- Collaborated with club executives to revamp web app and SEO capabilities for the 50+ member club.
- Improved page interactions by 10% and created a more accessible layout and UX for all audiences.
- Managed minor budget of \$1000 provided by the team for various improvement tasks on the web application.

RESEARCH / PROJECTS

BEM Tutoring & Consulting | [Deployment](#) Oct 2022 | May 2024

- Designed and built a staff and client management system with Supabase, React/Node/Express, OpenAI.
- Architected a scalable solution to account for a team of 15 workers and 150+ students and parents.

Digital Sign-in/out System | [GitHub](#) Oct 2022 | May 2024

- Developed an end-to end sign-in/out system using MySQL, Raspi, and React/Node.Js/Express.js.
- Integrated with 100+ students/admin and used as a production app (access restricted) at MA Academy.

"Dimensionally reduced model for predicting octanol-water partition coeff." | WPI Feb 2023 | August 2023

- Collaborated with the WPI Dynamics Lab to provide a user-friendly application component for a novel machine learning application published in the *Journal of Cheminformatics*.

Modeling T1 Resting-State MRI Variants in OCD Patients | [GitHub](#) | [Paper](#) Aug 2022 | Feb 2023

- Researched and developed novel neural networks (published) with strong levels of accuracy to determine the casual factor in OCD. Used TensorFlow, ResNet50, heatmap matrices, and deployment technologies.

CERTIFICATES / AWARDS

2025 WashU Meta Hackathon – Second Prize Mar. 2025
2024 GoatHacks (HackWPI) – Grand Prize Winner, Feb. 2024
State of Massachusetts - Seal of Biliteracy in English and Japanese Jan. 2024
Purple Comet International Math Meet – Honorable Mention (27th / 1100 participants) May, 2023

SKILLS

Languages: English (Native Proficiency), Tamil (Conversational), Japanese (Intermediate), and Spanish (elementary)
Programming: Python, C, C++, Java, PHP, JavaScript/Typescript, Dart
Databases: MySQL, MongoDB, Firebase Firestore, Supabase
Libraries: Socket.io, Redux, NumPy, TensorFlow, Sci-kit Learn, ShadCN / Tailwind
Frameworks: React.js, Next.js, Vue.js, Node.js, Flutter, React Native, Spring Boot, Llama stack
Interests: tutoring, cooking, pickup basketball, working out